

Digital Material Passport

ID DMP-METAL-002 - Version 1.0.0
Issue Date 2025-05-14 - Certificate Type EN 10204 3.1

Manufacturer		Customer	Business Transaction	
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com		Engineering Solutions Ltd. Tech Park Way 45 Cardiff CF14 5DU GB procurement@engisolutions.example.com	Order	PO-78902 · 2000 kg Pos. 10 · 2025-04-21
			Delivery	DN-56790 · 2000 kg Pos. 1 · 2025-05-13
Subcustomer American Heavy Industries Inc. 5000 Industrial Blvd Building C Detroit, MI 48201 US materials@heavyind.example.com		Goods Receiver Canadian Construction Corp 777 Construction Ave Toronto, ON M5H 2N2 CA logistics@canconstruct.example.ca		

Product

Product Name	Structural Steel S420N	Name (EN)	1.8902
Batch ID	H-10988-01	Shape	Plate - Width 2000 mm - Thickness 25 mm - Length 6000 mm
Surface Condition	Hot-rolled	Product Norm	EN 10025-3 (2019)
Weight	2000 kg		
Production Date	2025-05-10		
Country of Origin	DE		

Heat Treatment

Process: **Normalizing** - Lot: NORM-2024-0823-A12 - Furnace: NF-LINE-02 - Date: 2024-08-23

Stage	Temperature	Duration	Cooling	Atmosphere
Austenitizing	920 C	60 min		Air
Cooling	20 C		Air	

Chemical Analysis

Heat Number **H-10988** - Melting Process **EAF+LF+VD** - Casting Method **ContinuousCasting** - Casting Date **2025-05-09** - Sample Location **Ladle**

Symbol	C	Mn	Si	P	S	CEV
Unit	%	%	%	%	%	%
Min						
Max	0.20	1.60	0.50	0.025	0.015	0.44
Actual	0.16	1.48	0.28	0.016	0.010	0.41

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.41%

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status			
Tensile Strength					EN ISO 6892-1	✓			
Test temperature: 20°C									
Specimen: 1/3R, L									
No.	1	2	3	Average	Min	Max	Std Dev	Spec Min	Spec Max
Value [MPa]	558	562	560	560	558	562	2	520	680
Yield Strength					EN ISO 6892-1	✓			
Test temperature: 20°C, 3 specimens tested									
No.	1	2	3	Average	Min	Max	Std Dev	Spec Min	
Value [MPa]	442	445	448	445	442	448	3	420	
Elongation after fracture					EN ISO 6892-1	✓			
Gauge length 5.65#S ₀ , 3 specimens tested									
No.	1	2	3	Average	Min	Max	Std Dev	Spec Min	
Value [%]	23	24	25	24	23	25	1	19	
Reduction of Area¹					EN ISO 6892-1	✓			
No.	1	2	3	Average	Min	Max	Std Dev	Spec Min	
Value [%]	60	62	64	62	60	64	2	50	

Charpy V-notch Impact Energy²

EN ISO 148-1

✓

Specimen: Top, T-L

No.	1	2	3	Average	Min	Max	Std Dev	Spec Min
Value [J]	56	58	60	58	56	60	2	40

Charpy V-notch Impact Energy²

EN ISO 148-1

✓

Specimen: Bottom, T-L

No.	1	2	3	Average	Min	Max	Std Dev	Spec Min
Value [J]	54	57	59	56.7	54	59	2.5	40

Brinell Hardness

EN ISO 6506-1

✓

Ball: 10mm, Force: 3000kg, 5 indentations

No.	1	2	3	4	5	Average	Min	Max	Std Dev	Spec Min	Spec Max
Value [HBW]	183	185	187	184	186	185	183	187	1.58	150	220

Vickers Hardness

EN ISO 6507-1

✓

10 kg load, 5 indentations

No.	1	2	3	4	5	Average	Min	Max	Std Dev	Spec Min	Spec Max
Value [HV10]	192	195	198	194	196	195	192	198	2.35	160	230

Rockwell Hardness

HR

18 HRC

22 HRC

EN ISO 6508-1

✓

Elastic Modulus

E

210 GPa

EN ISO 6892-1

✓

Strain Hardening Exponent

n

0.18

ASTM E646

✓

Plastic Strain Ratio

r

1.2

1.0

EN ISO 10113

✓

0.2% Proof Strength¹

EN ISO 6892-1

✓

No.	1	2	3	Average	Min	Max	Std Dev	Spec Min
Value [MPa]	428	430	432	430	428	432	2	400

¹ 3 specimens tested ² Test temperature: -20°C

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10204:2004 type 3.1 and the specified standards. The results comply with the requirements.

Validated by **Johann Weber**, Quality Inspector, Quality Assurance - 2025-05-14